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Editors
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Dear Editors,

Please consider the manuscript *Generic Identifiability of Strongly Tree-Child Level-2 Jukes–Cantor Networks*.

The paper proves that, on the open Jukes–Cantor parameter domain, a generic exact site-pattern distribution determines any binary standard semi-directed strongly tree-child level-2 network, up to ordinary triangle redirection. The result classifies symmetric full-dimensional overlap and one-sided generic stochastic containment. Its reconstruction output is structural modulo triangle redirection; it does not assert that one fixed input distribution belongs to every redirected stochastic model. It is sharp: weakening the class to semi-directed topologies that merely admit one tree-child rooting permits a non-triangle ambiguity for every number of leaves at least four.

A short structural theorem first shows that every weakly tree-child binary level-2 topology automatically has at most one triangle per blob. The main statistical advance beyond the triangle-free level-2 theorem is then a local-to-global analysis that accommodates every triangle configuration possible in the strong class. The proof combines pointwise cut-split recovery, analytic extraction of local port tensors, bounded-support reconstruction of cycle and theta blobs, and exact finite atlases. The Jukes–Cantor trinet polynomial used in the final residual cases is due to Englander et al.; the new step is its application to restored support labels in all 192 seven-port core-3 completions.

The submission includes complete LaTeX and TikZ source and a regenerative computational supplement. Primitive generators rebuild the bounded graph and tensor universes; exact verifiers check polynomial identities, signs, and Jacobian minors; separately implemented reviewers replay the load-bearing calculations. Successful clean quick and full verification transcripts are included, and hashes are used only as integrity controls.

The manuscript is not under consideration elsewhere. No empirical biological data were used. I am the sole author, declare no competing interests, and accept responsibility for every theorem, citation, and artifact. The manuscript transparently discloses the AI systems used during exploration, code generation, drafting, and adversarial review; no AI system is listed as an author.

I believe the manuscript is well suited to readers interested in mathematical biology, phylogenetic-network inference, algebraic statistics, and identifiability. Thank you for your consideration.

Sincerely,

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